

Artificial Intelligence & Machine Learning - Task 2

Feature Engineering, Model Optimization & Performance Comparison

1. Introduction

The objective of this project is to develop an enhanced house price prediction system using machine learning. The project uses the California Housing Dataset, which contains 20,640 observations and eight input features related to housing, population and geographical location. The target variable is HousePrice, representing the median house value.

2. Methodology

The dataset was loaded using scikit-learn and converted into a pandas DataFrame. The target variable, HousePrice, was separated from the eight predictor features. The feature matrix was standardized using StandardScaler. The data was then divided into training and testing sets using an 80:20 split with `random_state = 42`. The resulting training set contained 16,512 observations and the test set contained 4,128 observations. Three regression algorithms were trained on the training data and evaluated on the unseen test data.

3. Models Used

Model	Purpose
Linear Regression	Used as a baseline regression model.
Ridge Regression	Linear regression with regularization to help reduce overfitting.
Decision Tree Regressor	Captures non-linear relationships between features and house prices.

4. Results

Model performance was evaluated using Root Mean Squared Error (RMSE) and R^2 score. A lower RMSE indicates lower prediction error, while a higher R^2 indicates stronger explanatory performance.

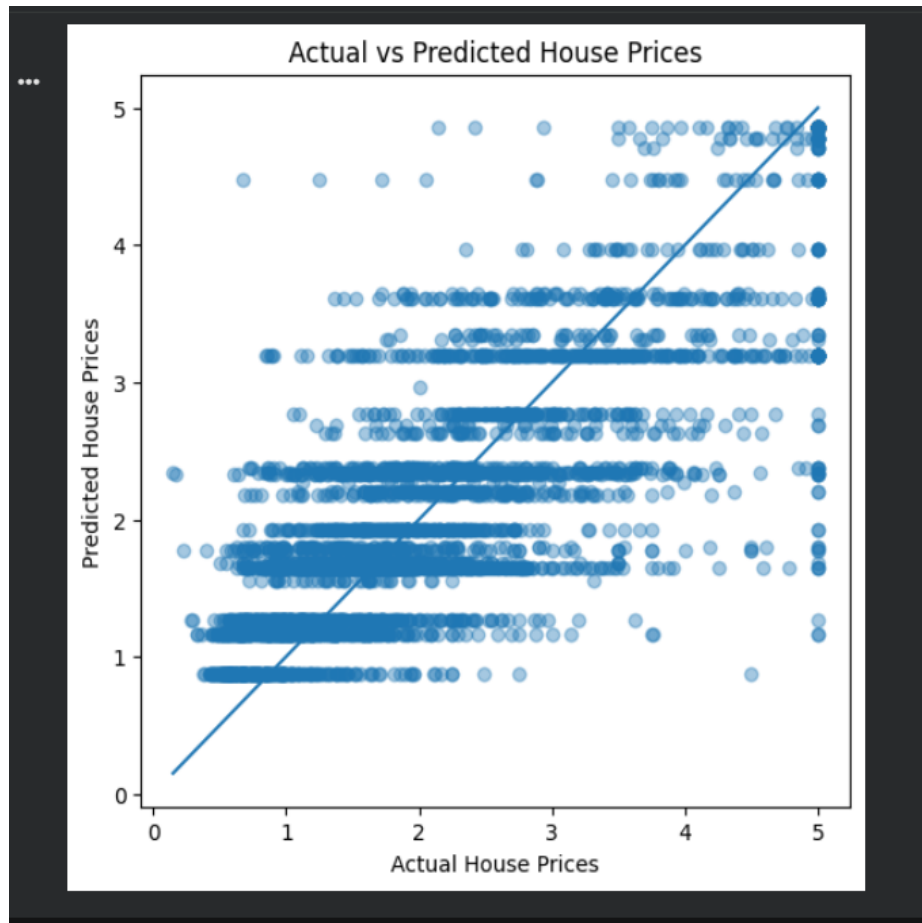
Model	RMSE	R^2 Score
Linear Regression	0.745581	0.575788
Ridge Regression	0.745554	0.575819
Decision Tree	0.724234	0.599732

5. Model Selection

The Decision Tree Regressor produced the lowest RMSE (0.724234) and the highest R^2 score (0.599732) among the three tested models. Ridge Regression was a very close second, while Linear Regression produced the lowest R^2 score. Based on both evaluation metrics, the Decision Tree was selected as the best-performing model for this experiment.

6. Visual Performance Validation

The Actual vs Predicted House Prices plot compares the observed house-price values with the predictions produced by the selected model. Points closer to the diagonal reference line indicate closer agreement between actual and predicted values.



7. Conclusion

The experiment demonstrated a complete machine learning workflow involving dataset preparation, feature scaling, train-test splitting, model training and quantitative performance comparison. Among the three models, Decision Tree Regressor achieved the best performance with an RMSE of 0.724234 and an R^2 score of 0.599732. Therefore, the Decision Tree model was selected as the final model for this Task 2 house price prediction experiment.

Dataset: California Housing Dataset | Training samples: 16,512 | Test samples: 4,128